AI-101 Mini-Project, Team-2

Online Shopper Purchase Prediction

Can we predict whether a visitor will purchase?

LOGISTIC REGRESSION

PROBLEM and DATASET

A visit is not the same as a purchase

We use session behavior to classify each online-shop visit as Purchase or No Purchase.

THE PREDICTION TASK

Input Pages viewed • time spent • month • visitor type • device/browser categories

Model Logistic regression

Output Yes / No

12,330
SESSIONS

17 + 1
FEATURES + TARGET

15.5%
PURCHASE RATE

CLASS IMBALANCE

No purchase:	**84.5%**
Purchase:	**15.5%**

A majority-class baseline already reaches 84.5% accuracy.

METHOD

From raw sessions to a trained classifier

The refined pipeline was guided by exploratory analysis and designed to avoid test-data leakage.

01

Explore

Summary statistics
Correlation heatmap

→

02

Prepare

Remove duplicates
Encode & group categories

→

03

Transform

log(1 + x)
Standardize features

→

04

Train

80/20 stratified split
Balanced logistic regression

→

05

Evaluate

Precision, recall, F1
ROC-AUC & confusion matrix

Key Choices: **Stratification preserves buyer share • scaling is fit on training data only • class_weight = 'balanced' gives the minority class more influence**

RESULTS

The refined model improves the first baseline

Test-set performance; the final model catches more purchases while keeping a similar train/test profile.

TEST-SET COMPARISON

Metric	First model	Final model
Accuracy	0.846	**0.869**
Precision	0.506	**0.556**
Recall	0.796	**0.814**
F1	0.619	**0.661**
ROC-AUC	0.908	**0.925**

TRAIN-SET VS TEST-SET

Metric	Training set	Testing set
Accuracy	0.869	**0.869**
Precision	0.556	**0.556**
Recall	0.792	**0.814**
F1	0.653	**0.661**
ROC-AUC	0.920	**0.925**

WHAT STANDS OUT

81% *of actual purchases are found (recall).*

56% *of predicted purchases are correct (precision).*

TAKEAWAYS

Useful signal — with important caveats

A strong ranking score does not mean every individual prediction is correct.

WHAT THE MODEL SUGGESTS

PageValues (OR ≈ 5.0): strongest positive association in the fitted model.

ExitRates (OR ≈ 0.60): higher exit rates are associated with lower purchase odds.

Train vs test: similar scores suggest limited overfitting on this split.

Precision vs recall: balanced weights help find buyers, but create some false alarms.

LIMITATIONS

Linear decision structure: may miss complex patterns.

PageValues: partly derived from purchase-related behaviour; interpret cautiously.

One split only: metrics may vary with another train/test split.

Anonymous codes: we cannot assign real-world meaning to every category.

TEAM 2

Sanjay Deep — **Documentation** | Keshav Garg — **Video** | Pratyush Rao Elchuri — **Slides** | Rudra Parde — **Execution**

NOTE: All members contributed code. AI-use disclosure: Claude assisted with explanations, analysis, interpretation and README formatting; code and results were run and checked by the team. ChatGPT assisted in PPT.